

Predicting Student Academic Performance

Team: Sirius

Manibai Erkenur (24B130697) | Yerdaly Yeskendir (23B031285) | Knyazev Artyom (24B031851)
Dataset: Student Performance Dataset | Problem Type: Regression | Target: G3 (Final Grade)

Section 1 — Abstract

This report presents a regression analysis aimed at predicting the final grade (G3) of students enrolled in a Portuguese language course. The dataset, sourced from the UCI Machine Learning Repository, contains 649 student records and 33 features covering academic history, demographic background, and social factors. Four regression models were trained and evaluated: Linear Regression (baseline), K-Nearest Neighbors (KNN), Decision Tree, and Support Vector Regression (SVR). All models were assessed using 5-fold cross-validation and held-out test set metrics (R^2 , RMSE, MAE). **Linear Regression achieved the best performance with Test $R^2 = 0.8633$, RMSE = 1.155, and MAE = 0.746.** The most important finding is that prior academic performance — specifically the second-period grade (G2) — is by far the strongest predictor of final grade, with a Pearson correlation of 0.92, confirming that grade trajectories within a single school year are highly stable and predominantly linear.

Section 2 — Introduction

The Student Performance Dataset was collected by Paulo Cortez and Alice Silva and is publicly available at the Machine learning repository

(<https://archive.ics.uci.edu/dataset/320/student+performance>).

The dataset records academic outcomes of 649 students from two Portuguese secondary schools in 2005–2006, all enrolled in Portuguese language courses. Each record covers 33 attributes including three period grades (G1, G2, G3), demographic variables (age, sex, address), family background (parental education and occupation, family size), study-related variables (study time, number of prior failures, school support, paid tutoring), and social indicators (going out frequency, alcohol consumption, internet access).

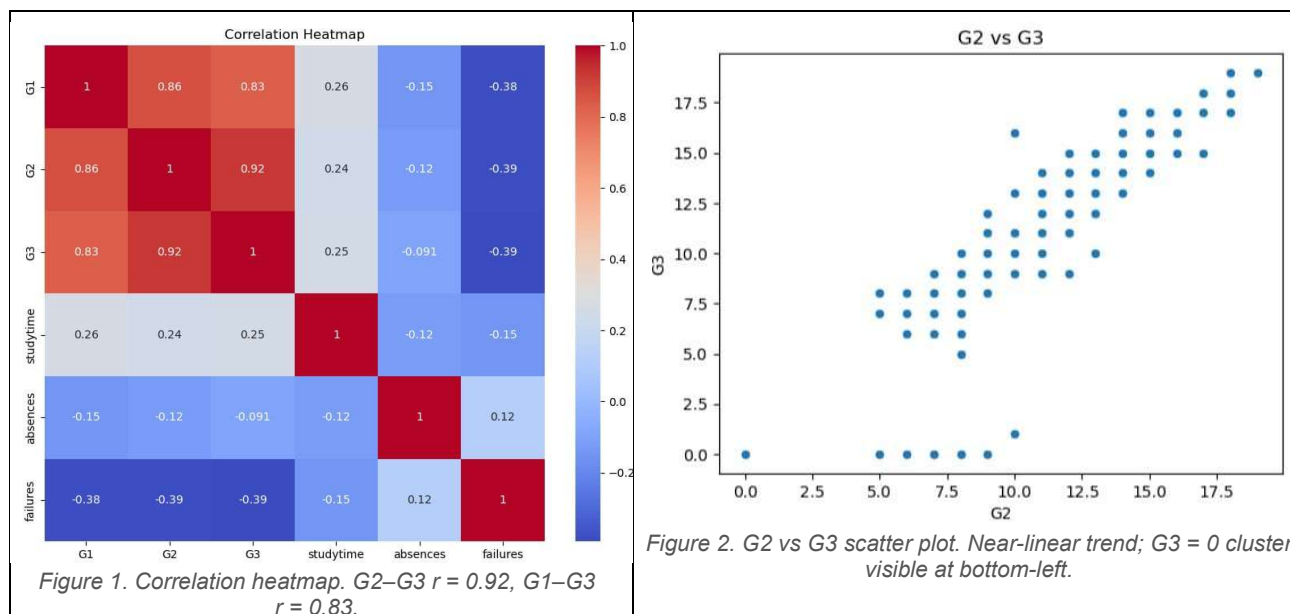
The task is a supervised regression problem. The target variable is G3, the student's final grade on a 0–20 integer scale. Predicting G3 accurately has direct practical value: early identification of at-risk students allows educational institutions to intervene before the end of the academic year.

The team's hypothesis, formulated in Week 1, was as follows: **previous performance (G1, G2), amount of study time, number of absences, and number of prior failures will all have a significant impact on students' final grade (G3).** We proposed G1, G2, studytime, absences, and failures as primary input features, with categorical variables to be encoded as additional inputs.

Section 3 — EDA Summary

Exploratory data analysis, conducted in Week 2 by Knyazev Artyom, revealed three key structural properties of the dataset that directly shaped all subsequent modelling decisions.

Finding 1 — Strong linear relationship between period grades and G3. The correlation heatmap (Figure 1) confirmed quantitatively what the scatter plots suggested visually: G2 has the strongest positive correlation with G3 at 0.92, followed by G1 at 0.83. The scatter plot of G2 vs G3 (Figure 2) makes this relationship particularly clear — the trend is nearly perfectly linear across the full grade range with little dispersion. Students who perform well in the second period almost invariably achieve high final grades. This finding directly informed the feature selection: G1 and G2 — together with the engineered features `avg_grade` and `grade_diff` — form the backbone of the predictive signal in all four models.



Finding 2 — The G3 = 0 cluster (withdrawn students). The histogram of final grades (Figure 3) shows G3 is approximately normally distributed with a mean near 11–12 and a slight negative skew. A distinct spike at G3 = 0 is clearly visible, representing roughly 8% of the sample (~50 students). These are not genuinely low-performing students — they withdrew from the course or were absent for the final exam. This structural anomaly creates a bimodal pattern in the outcome variable and poses a specific challenge for regression models, which cannot infer withdrawal intent from the available features.

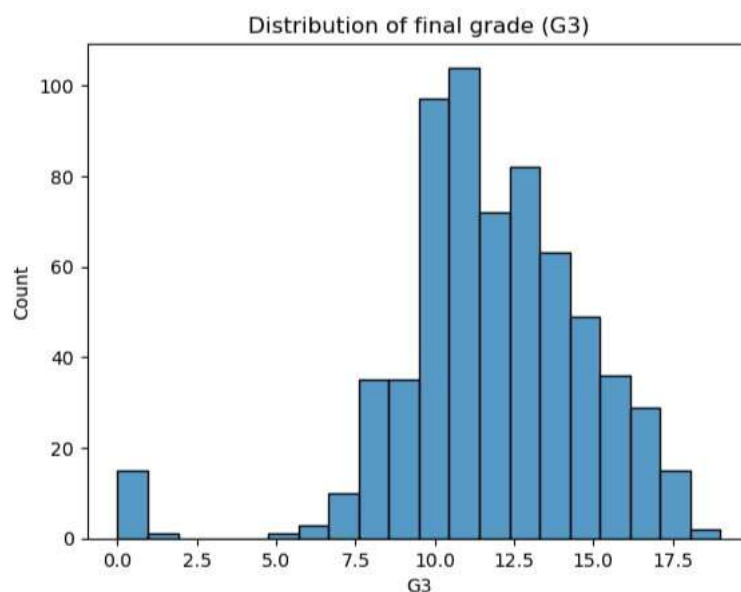


Figure 3. Distribution of final grade (G3). Approximately normal with a distinct spike at G3 = 0 representing withdrawn students (~8% of sample).

Finding 3 — Weak influence of study time and absences; moderate negative effect of failures. The heatmap assigned studytime a correlation of 0.25 with G3 and absences only -0.09 , confirming both as weak individual predictors. The failures variable showed a more meaningful negative correlation of -0.39 : box plots confirmed a clear downward trend in median G3 as prior failures increased. This partially supported the Week 1 hypothesis but with notably smaller effect sizes for studytime and absences than anticipated.

Section 4 — Methodology

Feature Set and Preprocessing

The final feature set comprised seven variables: G1, G2, studytime, absences, failures, avg_grade (mean of G1 and G2), and grade_diff (G2–G1). Categorical features (sex, address, Pstatus, and others) were tested with one-hot encoding; however, this produced less than 0.005 improvement in cross-validated R^2 , so no categorical encoding was applied in the final pipeline in order to maintain interpretability and avoid unnecessary dimensionality. The dataset contained no missing values, so no imputation was required.

The data was split 80/20 into training (519 samples) and test (130 samples) sets using `random_state=42` for reproducibility. `StandardScaler` was applied to normalise all features. The scaler was fitted exclusively on the training set and then applied to the test set, preventing any data leakage.

Models and Rationale

Linear Regression (baseline). Selected as the mandatory baseline. The near-linear G2–G3 relationship observed in the EDA made a linear model a well-motivated starting point. It requires no hyperparameter tuning and provides a clean benchmark for all other models.

K-Nearest Neighbors Regressor (KNN). Selected as a non-parametric, assumption-free model capable of capturing local grade patterns that a global linear fit may miss. The optimal k was chosen by sweeping k from 1 to 20 and selecting the value maximising mean 5-fold CV R^2 . Best $k = 8$.

Decision Tree Regressor. Selected because the $G3 = 0$ cluster is a natural early split point that a tree can isolate, potentially reducing the large residuals a linear model produces for withdrawn students. Max depth was tuned via 5-fold CV sweep over depths 1–15; best depth = 4. An unconstrained tree reached training $R^2 = 0.968$ but CV $R^2 = 0.677$ — a clear overfit demonstrating the necessity of depth regularisation.

Support Vector Regression (SVR, RBF kernel). Selected for its epsilon-insensitive loss, which provides robustness against outliers such as the $G3 = 0$ cluster. Default parameters ($C=1$, $\epsilon=0.1$) gave CV $R^2 = 0.717$; manual tuning to $C=10$, $\epsilon=0.5$ improved this to 0.799. An epsilon of 0.5 (half a grade point) better reflects the natural noise floor in student grading.

Cross-Validation

All models were evaluated using 5-fold cross-validation on the training set (`KFold`, `shuffle=True`, `random_state=42`). Three metrics were recorded per fold: R^2 , RMSE, and MAE. Results are reported as mean $\pm$ standard deviation across folds. Final test-set scores were computed once on the held-out 20% and reported alongside CV results for full transparency.

Section 5 — Results

Table 1 reports the full evaluation for all four models. Best test-set values per metric (all belonging to Linear Regression) are shown in bold.

Model	CV R^2 (mean $\pm$ std)	CV RMSE (mean $\pm$ std)	CV MAE (mean $\pm$ std)	Test R^2	Test RMSE	Test MAE
Linear Regression	0.841 $\pm$ 0.041	1.274 $\pm$ 0.273	0.817 $\pm$ 0.109	0.8633	1.155	0.746
KNN (k=8)	0.797 $\pm$ 0.037	1.444 $\pm$ 0.265	0.954 $\pm$ 0.124	0.8218	1.318	0.862
Decision Tree (depth=4)	0.818 $\pm$ 0.058	1.347 $\pm$ 0.302	0.853 $\pm$ 0.107	0.8153	1.342	0.751
SVR (RBF, C=10)	0.799 $\pm$ 0.062	1.428 $\pm$ 0.331	0.893 $\pm$ 0.145	0.8173	1.335	0.809

Table 1. Full model comparison across all metrics. CV scores are mean $\pm$ std over 5 folds. Best test-set values in bold.

Linear Regression is the best-performing model across all three test-set metrics. It achieves Test $R^2 = 0.8633$, explaining 86.3% of variance in final grades on unseen data. Its RMSE of 1.155

and MAE of 0.746 are the lowest among all models — on average, predictions are less than 0.75 grade points from the true value on a 0–20 scale. The margin over the next best model (KNN, $R^2 = 0.822$) is +0.041 in R^2 .

The cross-validation rankings mirror the test-set rankings exactly. Linear Regression leads at CV $R^2 = 0.841$ with the smallest fold-to-fold variance (std = 0.041), indicating the most stable generalisation. The Decision Tree and SVR show higher variance (std 0.058 and 0.062 respectively), suggesting greater sensitivity to data partitioning. All four models substantially outperform a naive mean-prediction baseline ($R^2 = 0$), confirming that each has extracted genuine predictive signal.

Section 6 — Error Analysis

Residual analysis across all four models is shown in Figure 4. Each column presents the actual vs predicted plot (top row) and the residuals vs fitted values plot (bottom row).

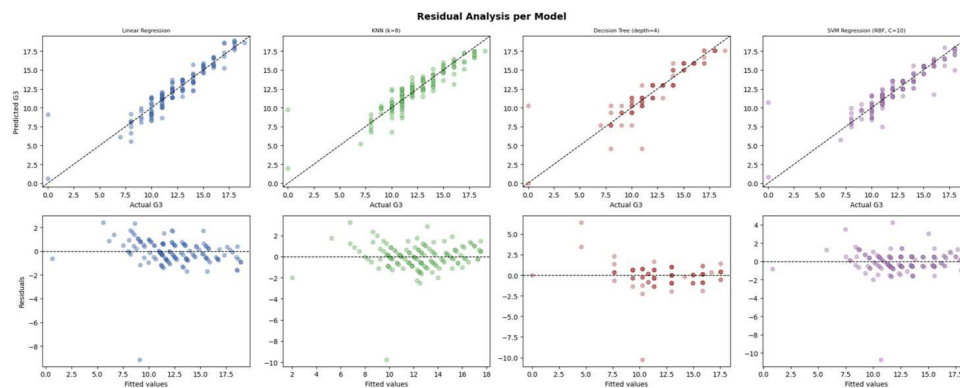


Figure 4. Residual analysis per model. Top row: Actual G3 vs Predicted G3 (dashed line = perfect prediction). Bottom row: Residuals vs fitted values. Patterns of systematic error are visible across models.

Pattern 1 — Systematic overprediction for withdrawn students ($G3 = 0$). The most consistently large residuals in the Linear Regression model (blue, far left) occur for students whose true $G3$ is 0. The model predicts these students at values between 6 and 9 — producing positive residuals of 6–9 grade points, visible as the cluster of points at the bottom-left of the actual vs predicted plot and as downward outliers in the residuals panel. These students had legitimate $G1$ and $G2$ grades in the 7–10 range, and the model correctly infers that such scores would ordinarily predict a mid-range outcome. The zero result is a withdrawal event, not an academic performance outcome, and cannot be predicted from the available features. The Decision Tree (red) partially reduces these errors by isolating the $G3 = 0$ cluster in an early split; the KNN (green) and SVR (purple) show similar outlier patterns.

Pattern 2 — Mild underprediction for top-performing students ($G3 = 18\text{--}20$). For students with actual $G3$ in the 18–20 range, all models tend to slightly underpredict by approximately 1–2 grade points. This is visible as a band of points above the dashed diagonal in the top-right corner of each actual vs predicted plot. It is a classic regression-toward-the-mean artefact: even students with $G2 = 17\text{--}18$ are pulled slightly downward by the global average structure of the fit.

Why complex models do not outperform the baseline overall. The Decision Tree reduces $G3 = 0$ residuals but produces higher errors across the dominant mid-range ($G3 = 7\text{--}15$), yielding a lower overall R^2 . KNN struggles with sparsity at the grade extremes in a 519-sample training set, and its residuals plot shows the largest spread of all models. SVR’s epsilon tube absorbs some outliers but does not improve materially on the main population. The linear model’s weakness on ~8% of the data is outweighed by its precision on the remaining 92%.

Section 7 — Reflection

Hypothesis revisited. The Week 1 hypothesis stated that $G1$, $G2$, studytime, absences, and failures would all significantly influence $G3$. This was partially confirmed but with an unexpected imbalance. $G1$ and $G2$ dominate entirely — $G2$ alone explains 84% of variance in $G3$ in a simple bivariate

regression. The failures variable contributed modestly ($r = -0.39$). Study time and absences proved far weaker than anticipated ($r = 0.25$ and -0.09 respectively); their removal did not materially affect model performance. The hypothesis correctly identified the direction of all effects but significantly overestimated the relative importance of study time and absences.

Most significant limitation. The $G3 = 0$ cluster is the most consequential unresolved problem. A proper treatment requires a two-stage model: first a binary classifier to predict whether a student will withdraw, then a separate regressor trained only on students who completed the course. This architecture would address both the large residuals for withdrawn students and the bias their inclusion introduces into the regression. Implementing this was beyond the current project scope.

What we would do differently. With more time, we would run a full GridSearchCV over SVR hyperparameters to ensure a global optimum rather than a manually explored point. We would also evaluate ensemble methods — Random Forest and Gradient Boosting — which can combine the linear grade signal with non-linear splits for the $G3 = 0$ boundary. Ordinal encoding for parental education (Medu, Fedu) may also capture meaningful structure currently discarded by the decision to exclude categorical features.

What additional data would help. A mid-year withdrawal flag — recorded at the point a student stops attending rather than inferred retrospectively from $G3 = 0$ — would allow the two subpopulations to be modelled separately. Teacher mid-term assessments would also likely improve predictions for students in the borderline $G3 = 6-9$ range where model uncertainty is highest.

References

Student	Performance	Dataset.	UCI	Machine	Learning	Repository.
https://archive.ics.uci.edu/dataset/320/student+performance						